

Astronomy 102 - Lab 11

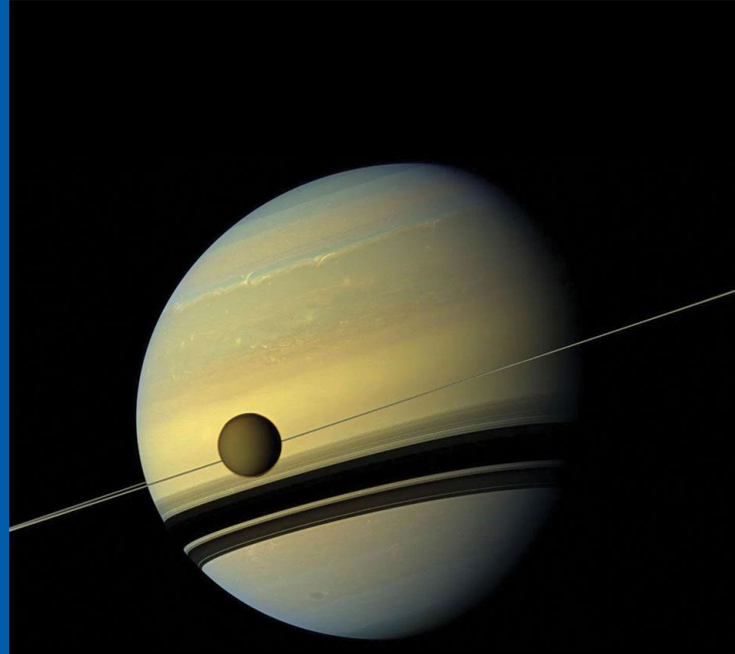
Black Holes

Matthew Prest

The latest in Astronomy

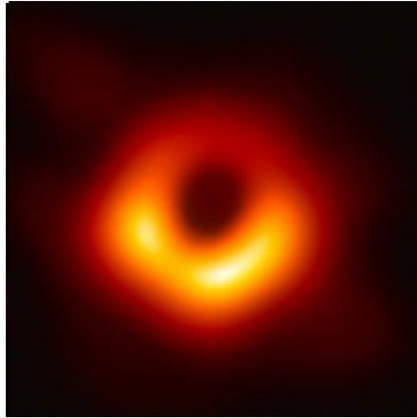
Saturn's largest moon Titan

- Second-largest moon in our solar system
- Christiaan Huygens discovered Saturn's largest moon, Titan, on March 25, 1655
- NASA's Dragonfly mission launch 2028
- Cassini-Huygens surface landing Jan. 14, 2005
- New ocean modeling results simulating the behaviour of Titan's liquid methane lakes and oceans

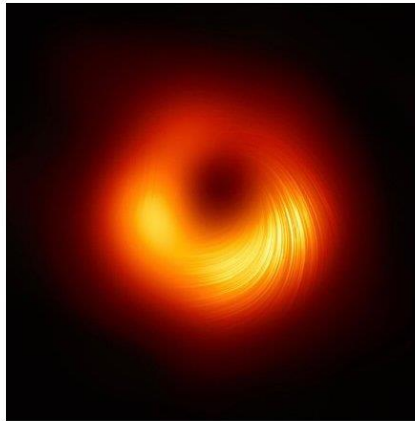


The latest in Astronomy

2019



2021

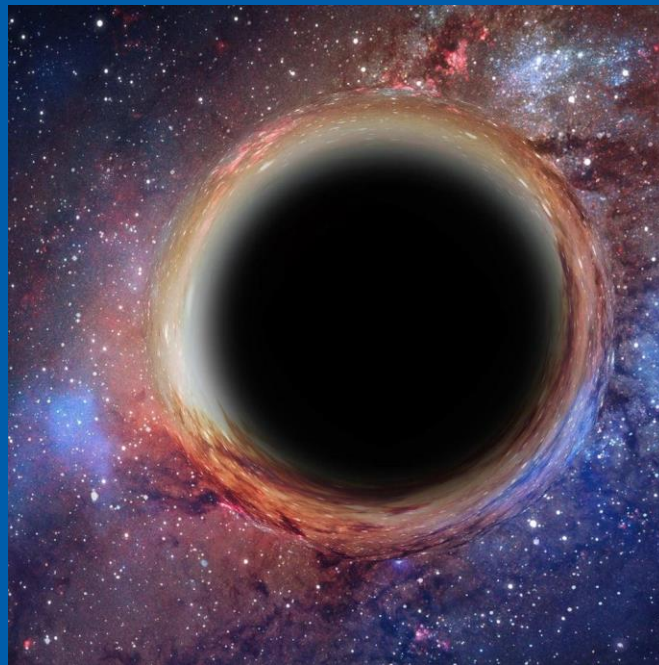
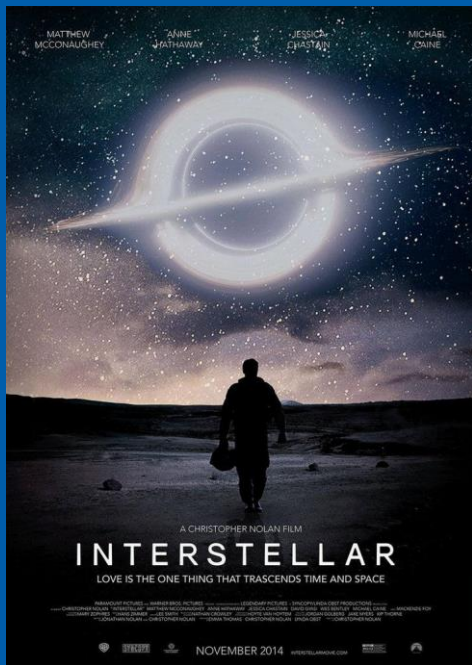


Simulated



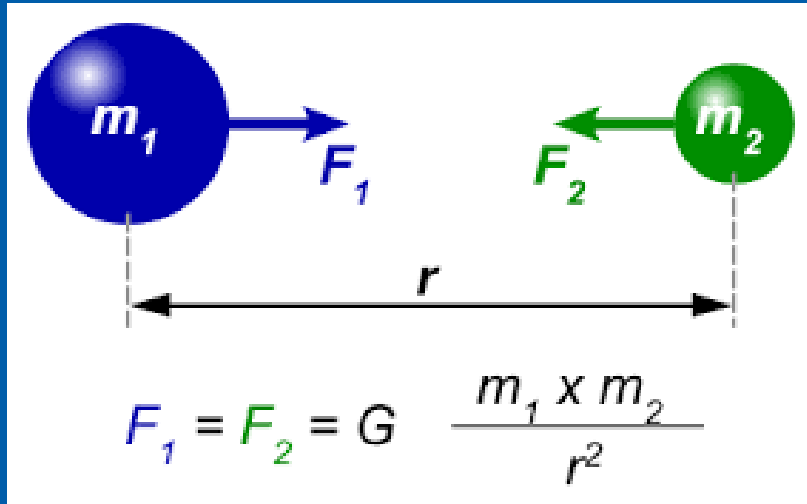
Dr. Katie Bouman's C.H.I.R.P Algorithm 2019

What is a Black Hole?



Remember Newton

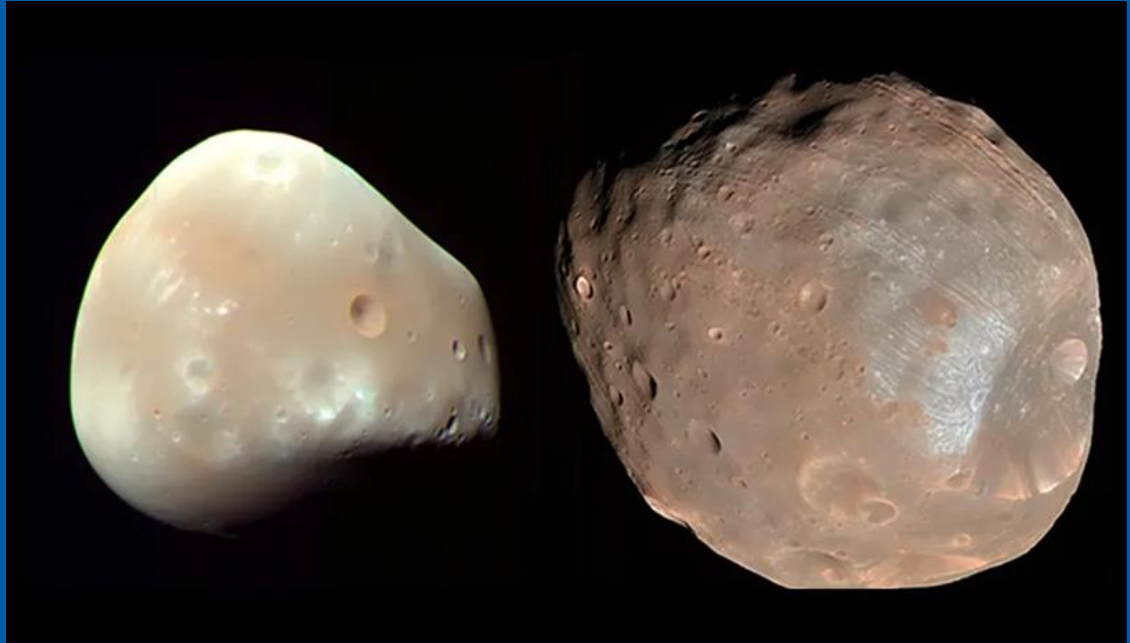
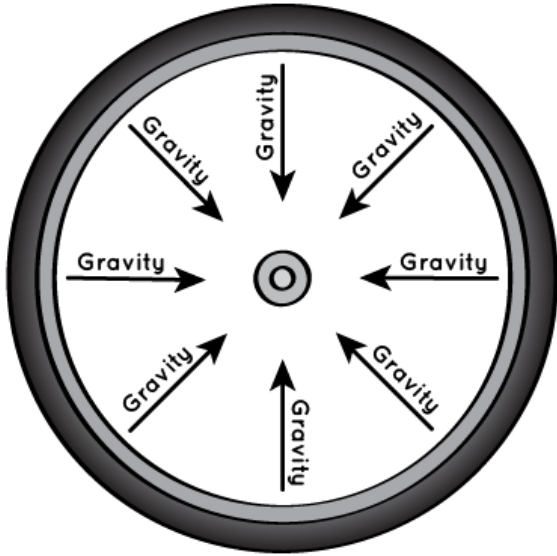
Mass attracts mass



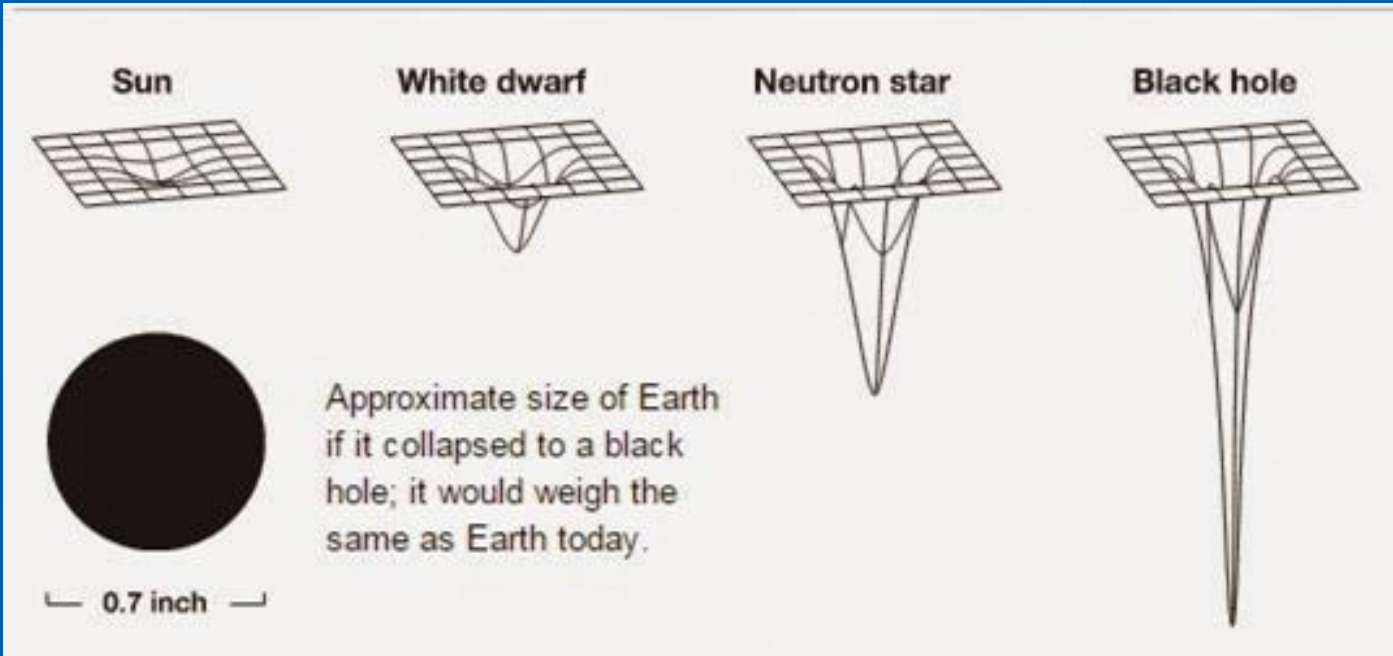
Remember Newton

Mass attracts mass

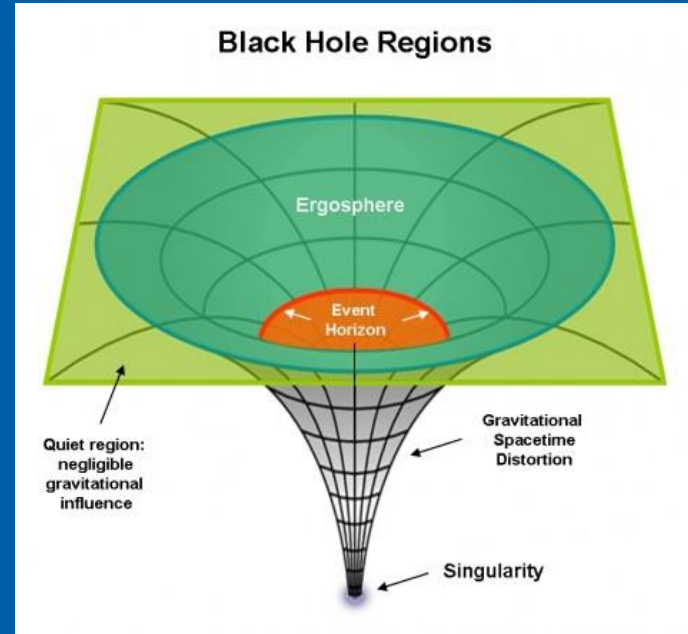
- Causes planets to round! (Provided they are big enough!)



What happens if we keep adding mass?

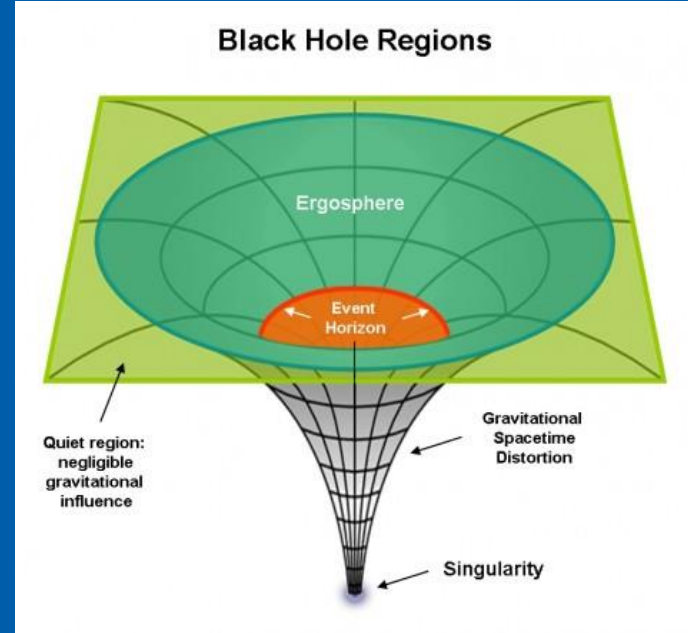
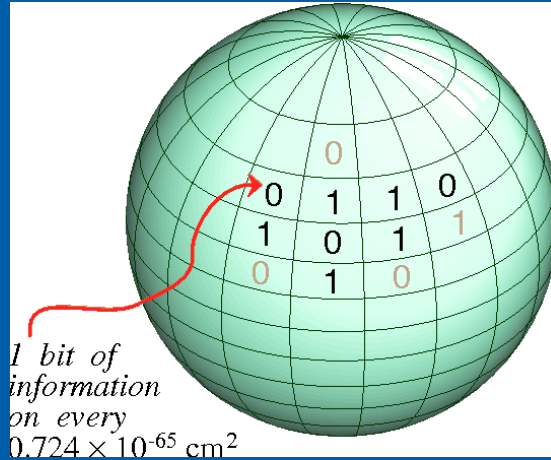


What are the properties of a Black Hole?



What are the properties of a Black Hole?

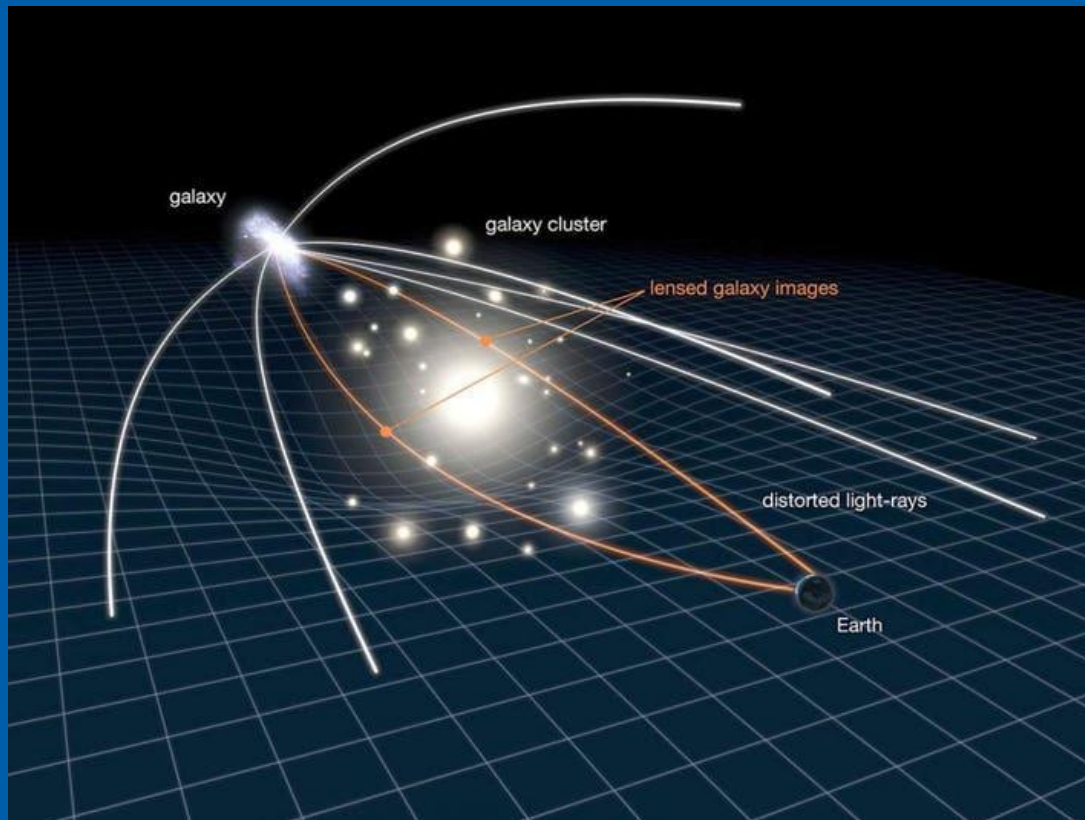
- Mass
- Radius
- Angular Momentum
- Charge
- Entropy



What about light?

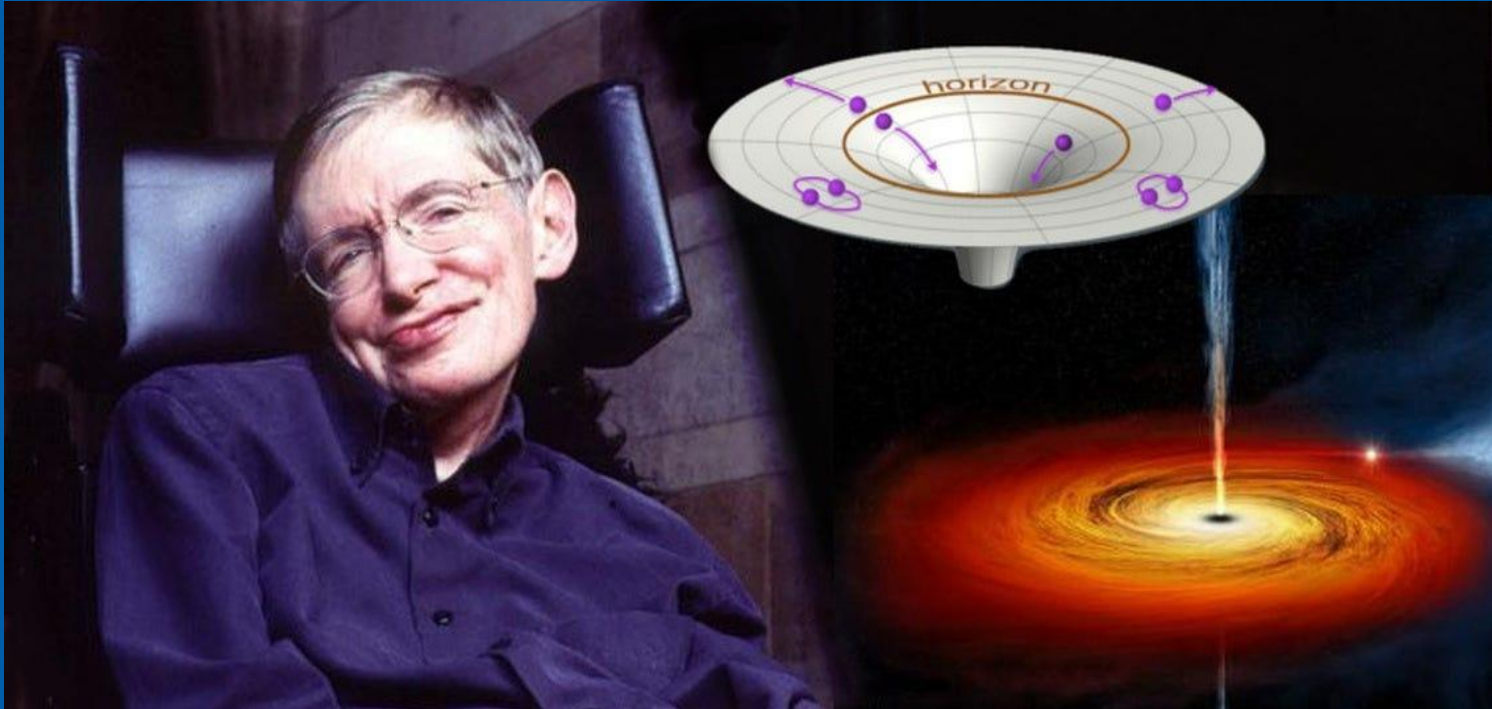
Remember light is massless!

- Yet we observe light “bending” due to gravity
- We call this gravitational lensing



Do they just keep growing?

Surprisingly no! Hawking radiation is produced that eventually shrinks the black hole.



Reminders

1. Complete the Tutorial first!

- This is best done now in class where I am available to assist

1. Attempt the essay questions last!

- These should be based on what you have covered in the tutorial.

1. Previous week's lab report due at the beginning of the next lab!

- Final grades are based on best 6 online labs plus each of the 4 in-class labs

1. Grades will be posted to Brightspace

- Check now that you have access to our course!
- Grade feedback will also be provided on Pearson, please check as this is the best way to improve your future grades!

1. Office hours

- Office hours are available on Zoom Mondays 4-6pm (same zoom link as class) as well as by appointment either in-person on or zoom. Come say hi, I'm here to help!

Any and all questions are welcome!